Inkplate 6

Inkplate 6: open-source 6" e-paper display with ESP32, WiFi/Bluetooth, ultra-low power, Arduino/MicroPython ready. Plug-and-play for hobbyists & makers.

If you're like us, the first time you saw an **e-reader**, you thought to yourself, "I could do something with that." Thanks to clean lines, high contrast, daylight readability, and the remarkable level of **energy efficiency** that comes from drawing power only when changing the contents of the screen, **e-paper** is uniquely suited to many applications. With **Inkplate 6**, our goal is to make e-paper accessible to hobbyists and DIY product designers by offering a plug-and-play hardware platform that is super-easy to use and compatible with **Arduino** and **MicroPython**.

To name a few features, Inkplate 6 has a stunning 6-inch e-paper display with a refresh rate of 1.26s, partial update 0.26s, greyscale mode, and partial updates support. Powered by **ESP32**, you will have a strong microcontroller with **WiFi** and **Bluetooth** at your disposal. It's super-low-power (22uA) so you can use it for days, weeks, or months out of a single battery charge. Using our **Arduino library**, it's up and running in 5 minutes. The **Micropython module** is available as well. It's 100% open-source for both software and hardware. A **USB-C to USB-C** cable is included in the package, so you can start using it right out of the box.

Designed for high-performance embedded development, the Inkplate 6 fully supports Espressif's official **ESP-IDF** framework. This native environment grants developers low-level control over the hardware, enabling precise memory management, optimized display rendering routines, and custom power profiling to take full advantage of the board's ultra-low deep sleep capabilities. Whether building home prototypes or commercial-grade e-paper applications, ESP-IDF provides the robust foundation needed to unlock the full potential of Inkplate's wireless and processing features.

Inkplate 6 options:

- **With e-paper:** The basic version with all you need to get started.
- **With e-paper display, enclosure, and battery:** Upgrade to this version if you're looking for both protection and mobility. This option includes a 3D-printed enclosure as well as an integrated battery, ensuring your device stays powered even on the go.

Choose the one that best fits your needs and enhance your experience!

Display

Display Size (inch)	6 inch
Display Resolution W	800
Display Resolution H	600
Display Color	Grayscale
Frontlight	No
Full Refresh (s)	1.26 s
Partial Refresh (s)	0.26 s
Touch Support	None

MCU

Mcu Part Number	ESP32 WROVER-E
Mcu Clock (MHz)	240 MHz

Connectivity

Wifi	Wi-Fi 4 (802.11n)
Bluetooth	Bluetooth Classic + BLE

Interface

Qwiic Compatible	Yes
Sd Card Slot	Yes

Power

Battery Included	No
Battery Connector	JST 2-pin connector
Sleep Current Ua	25

Other

Dev Environments	Arduino IDE, MicroPython, ESP-IDF, ESPHome / Home Assistant
Has Enclosure	No
Rtc Onboard	Yes

Variants

333232	With e-paper
333229	With e-paper, Enclosure & Battery

What's in the box

1× Inkplate 6
1× USB-C Cable

Typical applications

Inkplate 6 suits projects that show information for long periods without draining a battery. You can build a room sign that updates its schedule once an hour, a weather display that pulls data over Wi-Fi, or a shelf label that shows stock counts. Because the e-paper display only draws power when the image changes, a single battery charge can run a display for weeks. Use the SD card slot to show a rotating set of images without a network connection, or use the qwiic port to add a sensor and show live readings. The ESP32 microcontroller and Arduino or MicroPython support make it straightforward to fetch data from an API and render it as text or a bitmap.

Resources and downloads

Hardware Details	https://docs.soldered.com/inkplate/6/hardware/design/ (WEB)
Full Documentation	https://docs.soldered.com/inkplate/6/overview (DOCS)
Arduino Library	https://github.com/SolderedElectronics/Inkplate-Arduino-library (GITHUB)
Arduino: Get Started	https://docs.soldered.com/inkplate/6/quick-start-guide/ (DOCS)
Arduino: Get Started	https://inkplate.readthedocs.io/en/latest/get-started.html (DOCS)
MicroPython Module	https://github.com/SolderedElectronics/Inkplate-micropython (GITHUB)
OSHWA Certification	https://certification.oshwa.org/hr000003.html (PDF)